AviLight Ecological Impact and Diagnostics Report

Executive Summary

Run Context

Field	Value
Area	All Areas
Trend Range	2014 to 2025
Snapshot Range	2025-01 to 2025-01
Generated	2026-04-24 09:09:32 UTC

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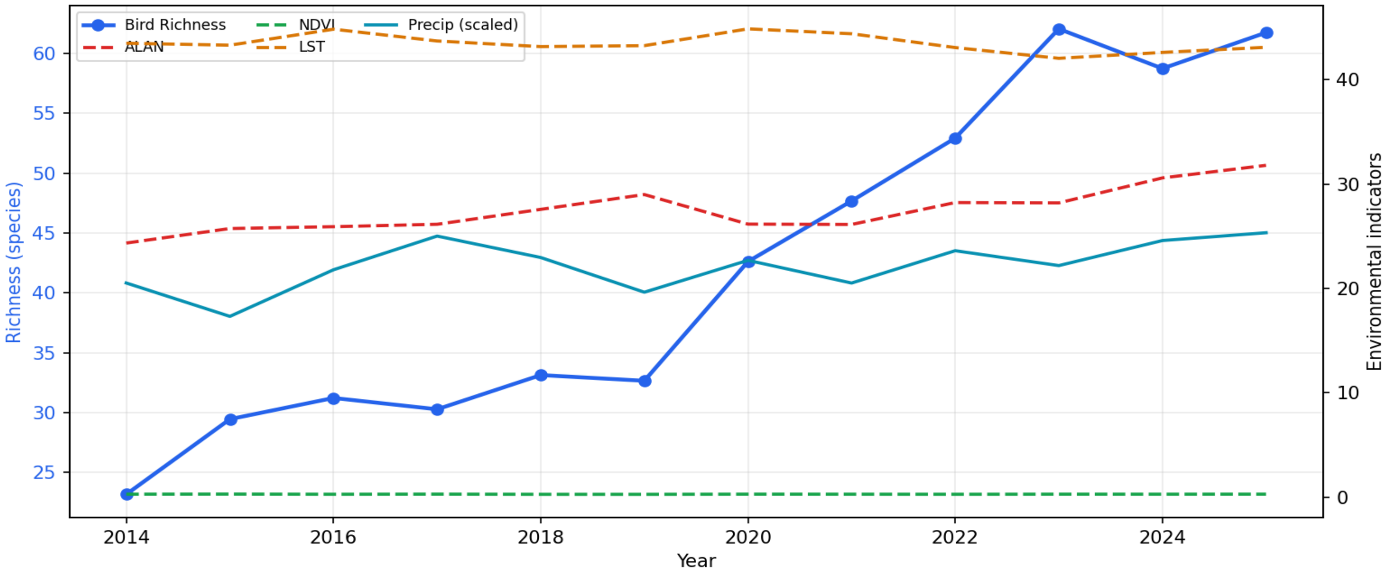
Ecological Impact View

Long-Term Ecological Impact Trends

Area: All Areas | Trend range: 2014 to 2025

Chart: Long-Term Ecological Impact Trends



Numerical Values: Historical Trends

Year	Richness	ALAN	NDVI	LST	Precip
2014	23.18	24.35	0.2770	43.51	2052.59
2015	29.46	25.75	0.2874	43.32	1732.52
2016	31.23	25.92	0.2711	44.86	2179.36
2017	30.29	26.15	0.2829	43.73	2502.38
2018	33.14	27.58	0.2659	43.19	2296.71
2019	32.67	29.01	0.2643	43.28	1964.26
2020	42.62	26.17	0.2801	44.89	2269.63
2021	47.69	26.14	0.2759	44.42	2052.01
2022	52.94	28.24	0.2698	43.09	2361.96
2023	62.06	28.21	0.2786	42.06	2219.41
2024	58.75	30.60	0.2762	42.63	2459.51
2025	61.75	31.81	0.2805	43.12	2534.40

Correlation Matrix

Area: All Areas | Trend range: 2014 to 2025

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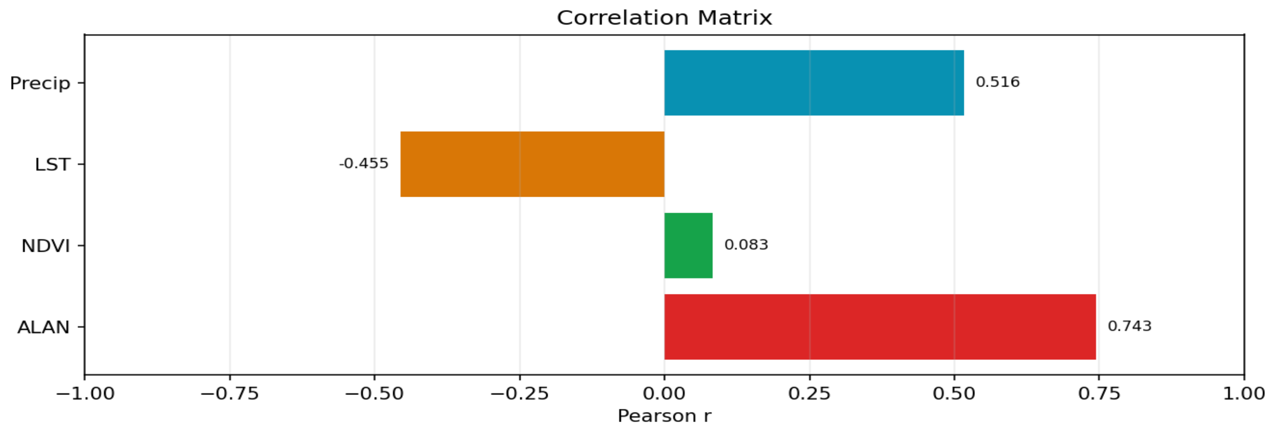
Chart: Correlation Matrix



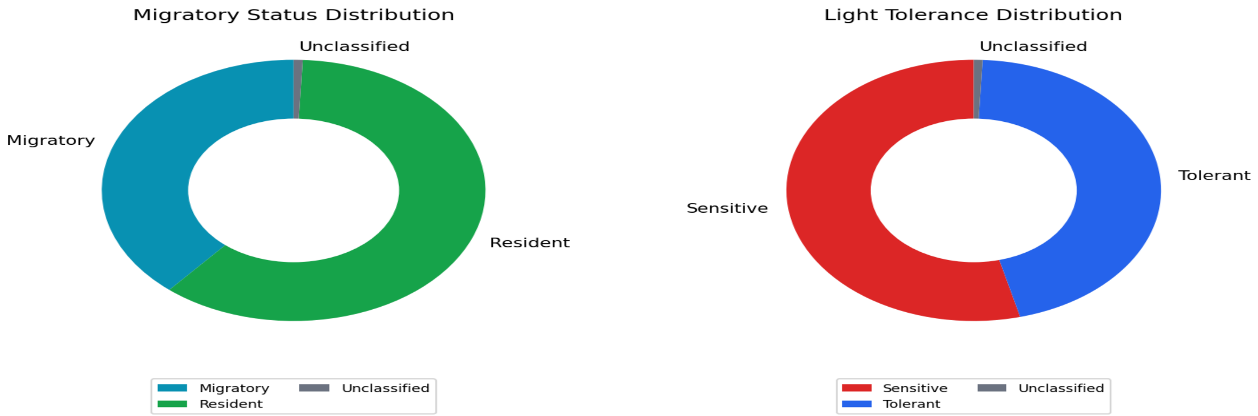
Numerical Values: Correlation Coefficients

Metric	Pearson r
ALAN vs Richness	0.7434
NDVI vs Richness	0.0829
LST vs Richness	-0.4549
Precipitation vs Richness	0.5159

Immediate Site Assessment & Vulnerability

Area: All Areas | Snapshot range: 2025-01 to 2025-01

Chart: Migratory and Light Tolerance Distribution



Numerical Values: Migration Distribution

Category	Count
Migratory	102
Resident	159
Unclassified	2

Numerical Values: Light Tolerance Distribution

Category	Count
Sensitive	142
Tolerant	119
Unclassified	2

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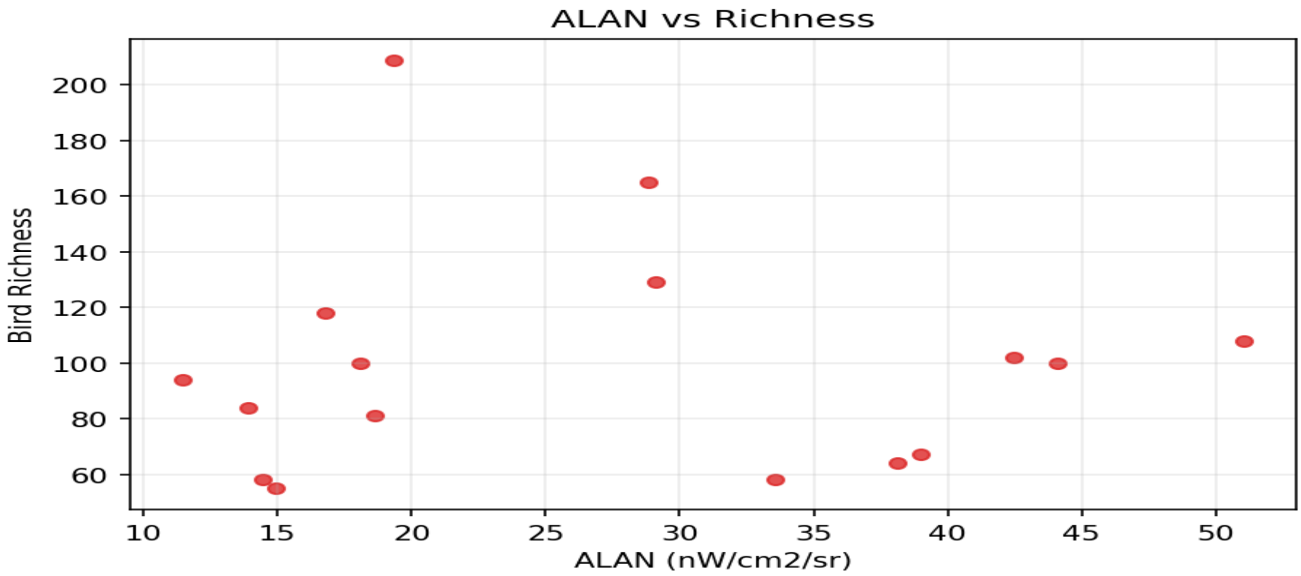
Chart: ALAN vs Richness



Numerical Values: ALAN vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	14.4463	58.00
Las Piñas	Las Piñas	16.7720	118.00
Makati	Makati	44.1109	100.00
Malabon	Malabon	14.9254	55.00
Mandaluyong	Mandaluyong	38.1263	64.00
Manila	Manila	42.4859	102.00
Marikina	Marikina	18.6270	81.00
Muntinlupa	Muntinlupa	18.0834	100.00
Navotas	Navotas	11.4724	94.00
Parañaque	Parañaque	28.8103	165.00
Pasay	Pasay	51.0255	108.00
Pasig	Pasig	33.5516	58.00
Quezon City	Quezon City	19.3597	209.00
San Juan	San Juan	38.9975	67.00
Taguig	Taguig	29.0985	129.00
Valenzuela	Valenzuela	13.8891	84.00

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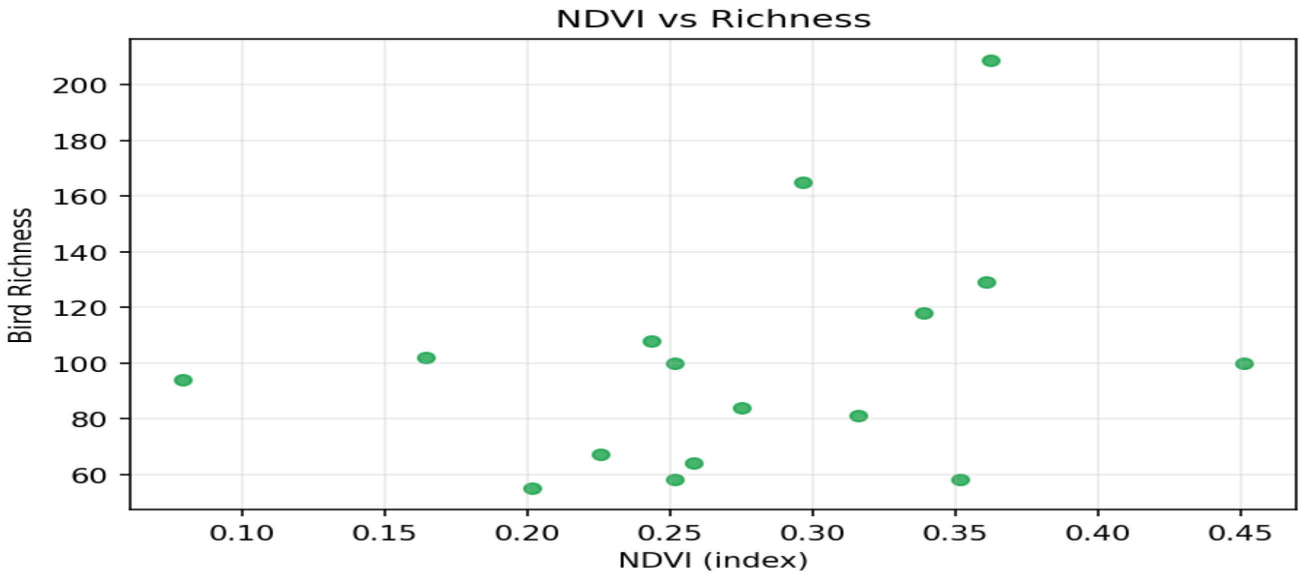
Chart: NDVI vs Richness



Numerical Values: NDVI vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	0.3517	58.00
Las Piñas	Las Piñas	0.3391	118.00
Makati	Makati	0.2515	100.00
Malabon	Malabon	0.2016	55.00
Mandaluyong	Mandaluyong	0.2585	64.00
Manila	Manila	0.1645	102.00
Marikina	Marikina	0.3158	81.00
Muntinlupa	Muntinlupa	0.4510	100.00
Navotas	Navotas	0.0793	94.00
Parañaque	Parañaque	0.2966	165.00
Pasay	Pasay	0.2437	108.00
Pasig	Pasig	0.2518	58.00
Quezon City	Quezon City	0.3626	209.00
San Juan	San Juan	0.2256	67.00
Taguig	Taguig	0.3606	129.00
Valenzuela	Valenzuela	0.2750	84.00

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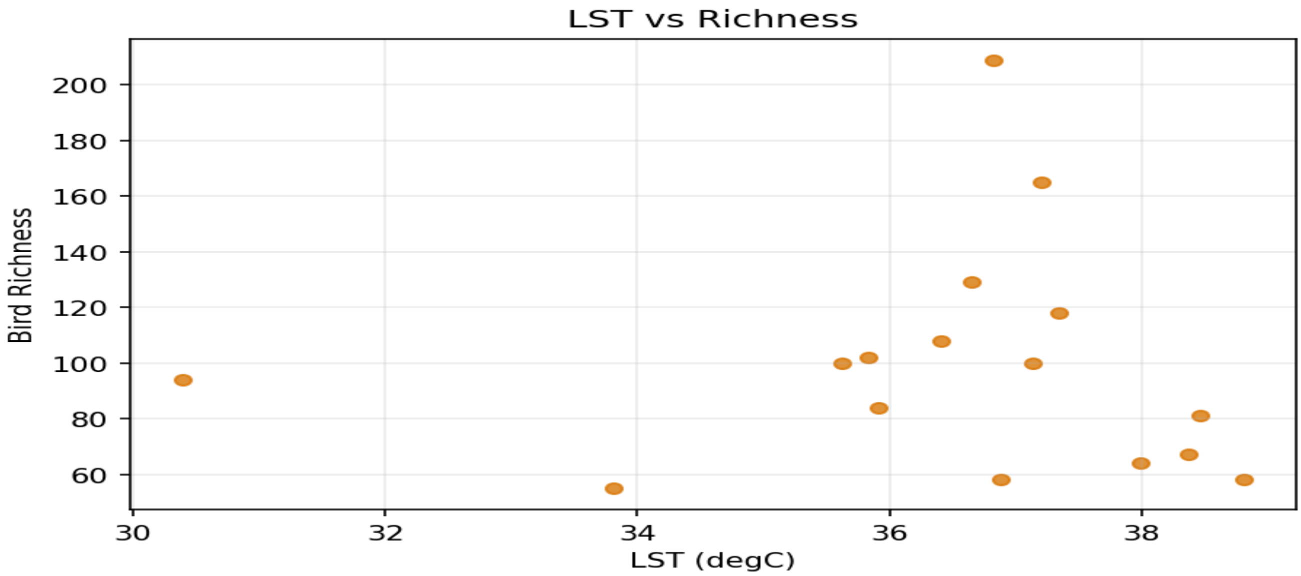
Chart: LST vs Richness



Numerical Values: LST vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	36.8790	58.00
Las Piñas	Las Piñas	37.3394	118.00
Makati	Makati	37.1364	100.00
Malabon	Malabon	33.8135	55.00
Mandaluyong	Mandaluyong	37.9907	64.00
Manila	Manila	35.8266	102.00
Marikina	Marikina	38.4591	81.00
Muntinlupa	Muntinlupa	35.6290	100.00
Navotas	Navotas	30.3941	94.00
Parañaque	Parañaque	37.2110	165.00
Pasay	Pasay	36.4101	108.00
Pasig	Pasig	38.8079	58.00
Quezon City	Quezon City	36.8196	209.00
San Juan	San Juan	38.3699	67.00
Taguig	Taguig	36.6470	129.00
Valenzuela	Valenzuela	35.9124	84.00

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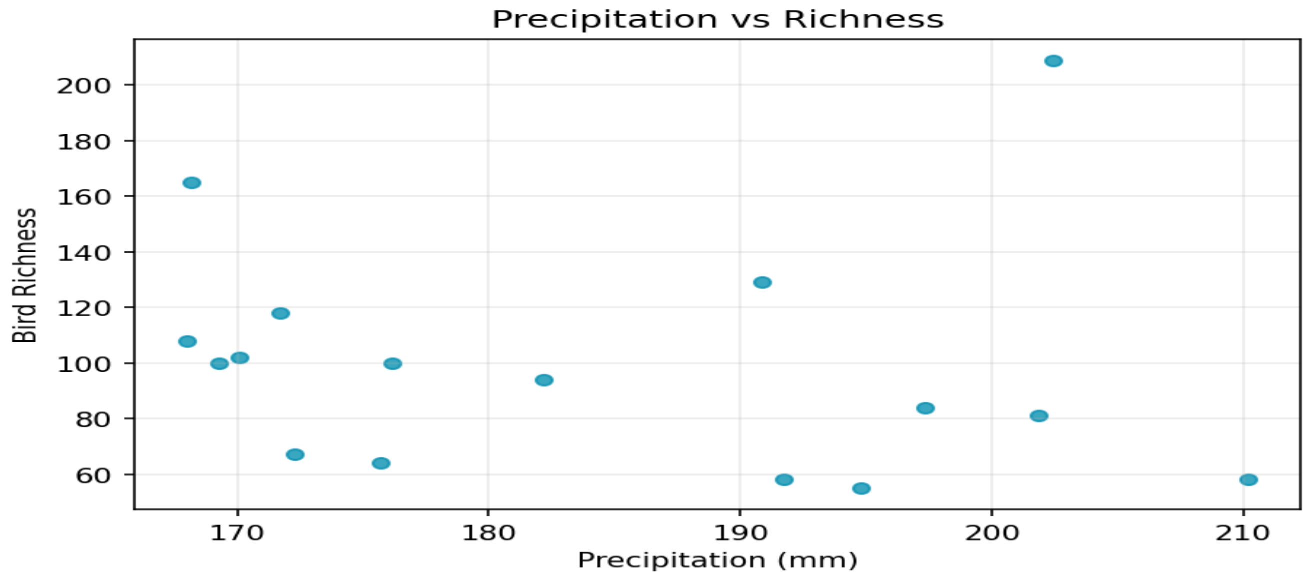
Chart: Precipitation vs Richness



Numerical Values: Precipitation vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	210.1734	58.00
Las Piñas	Las Piñas	171.6815	118.00
Makati	Makati	169.2433	100.00
Malabon	Malabon	194.7989	55.00
Mandaluyong	Mandaluyong	175.6956	64.00
Manila	Manila	170.0834	102.00
Marikina	Marikina	201.8672	81.00
Muntinlupa	Muntinlupa	176.1285	100.00
Navotas	Navotas	182.1797	94.00
Parañaque	Parañaque	168.1543	165.00
Pasay	Pasay	168.0031	108.00
Pasig	Pasig	191.7241	58.00
Quezon City	Quezon City	202.4204	209.00
San Juan	San Juan	172.2692	67.00
Taguig	Taguig	190.8491	129.00
Valenzuela	Valenzuela	197.3602	84.00

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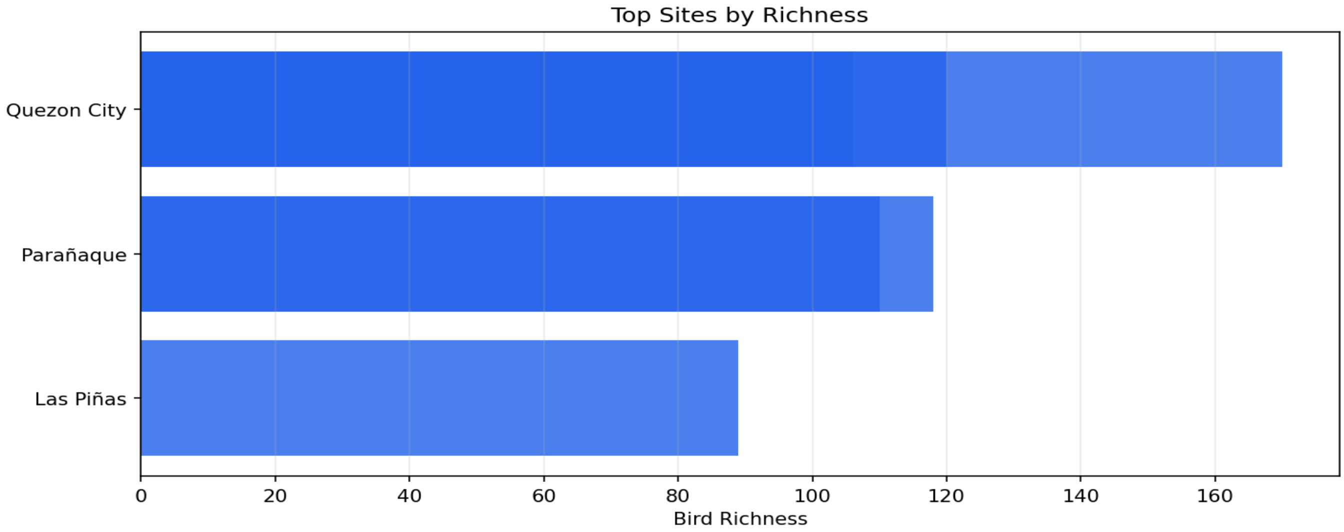
Chart: Top 10 Sites by Richness



Numerical Values: Top Sites by Richness

Rank	Area	Richness
1	Quezon City	170
2	Quezon City	120
3	Parañaque	118
4	Parañaque	110
5	Quezon City	106
6	Quezon City	100
7	Quezon City	95
8	Las Piñas	89
9	Quezon City	86
10	Quezon City	85

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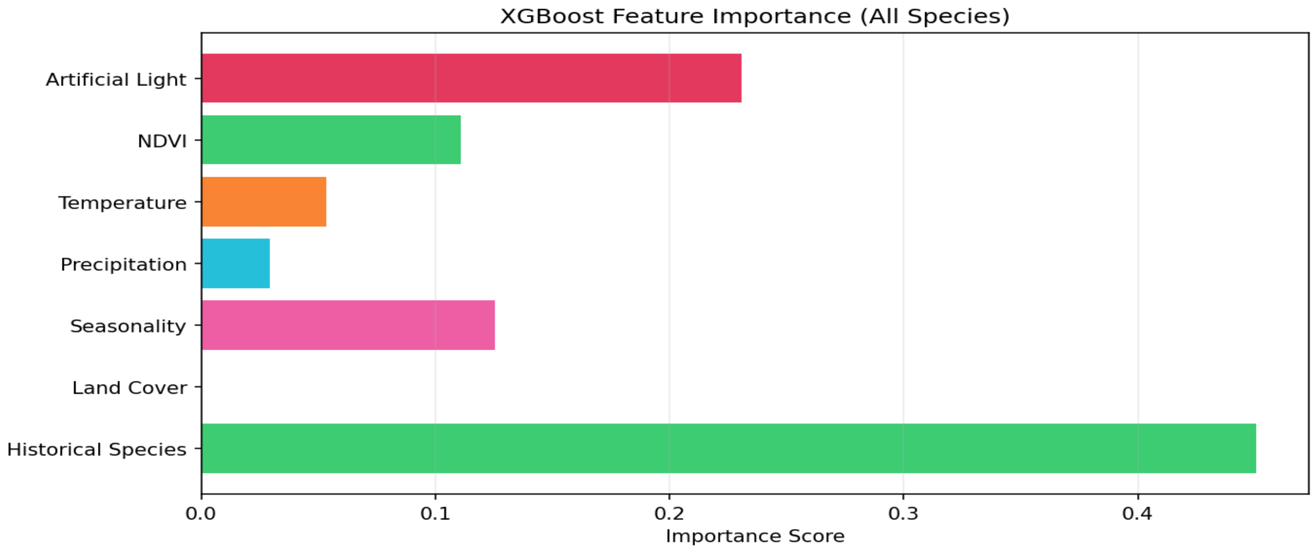
Model Diagnostics View

KPI Cards

Metric	Value
RMSE	6.3924
MAE	4.0568
R2	0.7445

Chart: XGBoost Feature Importance (All Species)



Numerical Values: XGBoost Feature Importance (All Species)

Feature	Importance
Artificial Light	0.230796
NDVI	0.110724
Temperature	0.053409
Precipitation	0.029266
Seasonality	0.125400
Land Cover	0.000000
Historical Species	0.450406

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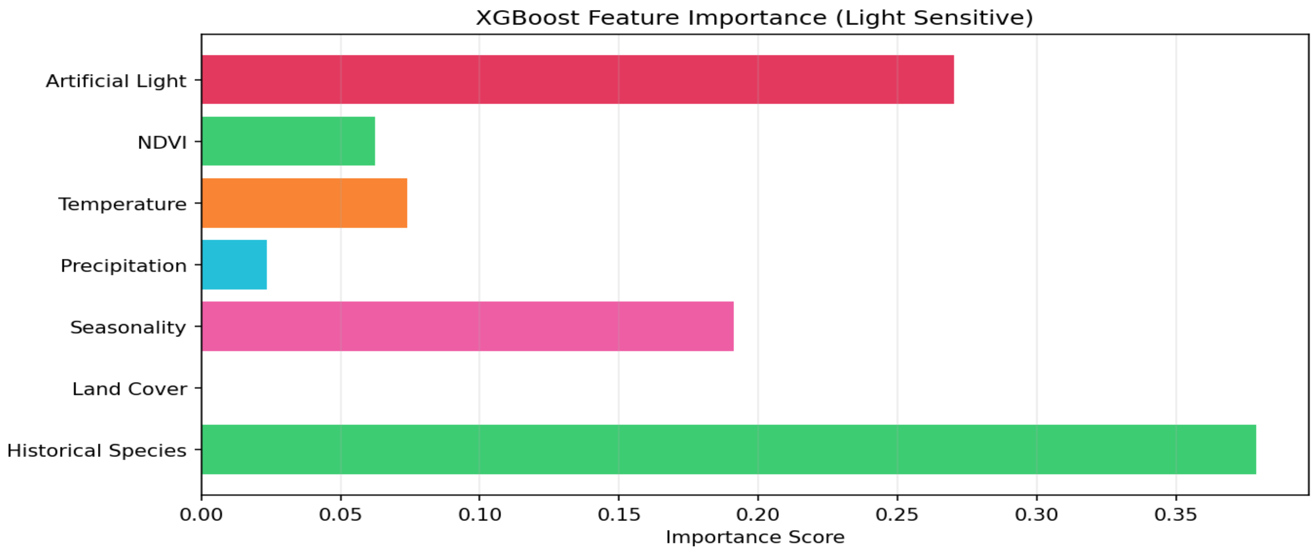
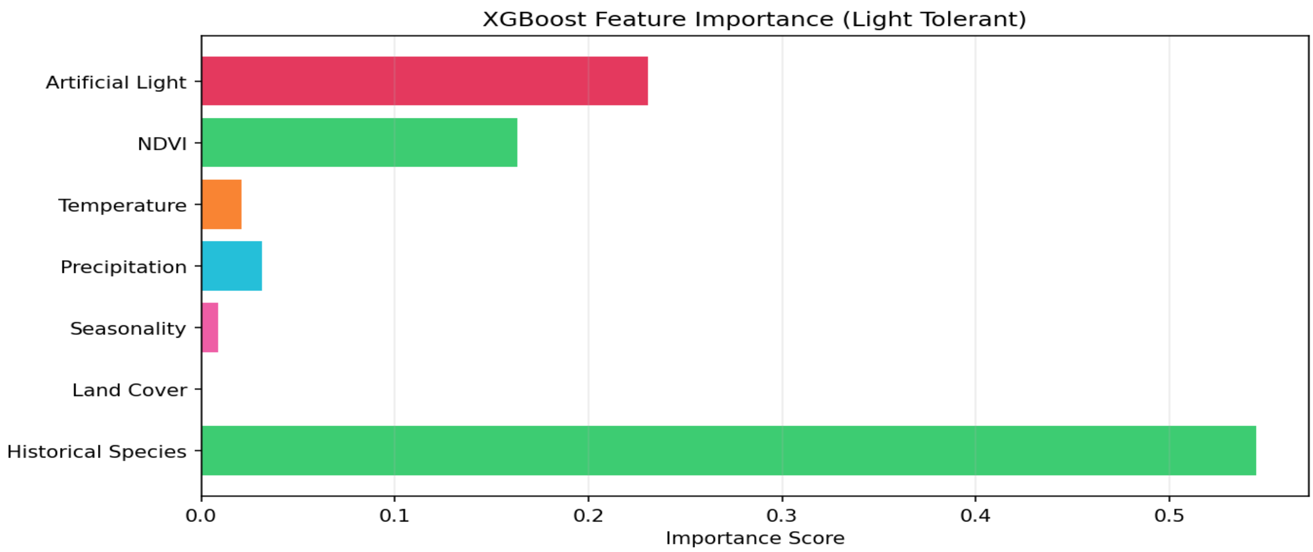
Chart: XGBoost Feature Importance (Light Sensitive)



Numerical Values: XGBoost Feature Importance (Light Sensitive)

Feature	Importance
Artificial Light	0.270363
NDVI	0.062340
Temperature	0.073999
Precipitation	0.023678
Seasonality	0.191019
Land Cover	0.000000
Historical Species	0.378602

Chart: XGBoost Feature Importance (Light Tolerant)

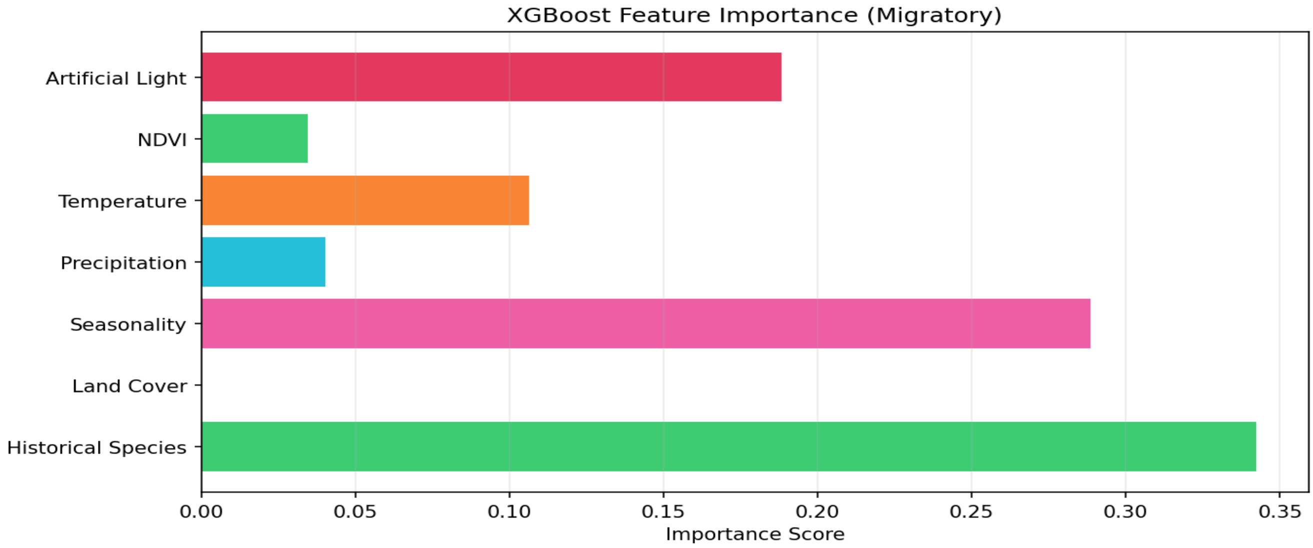


Numerical Values: XGBoost Feature Importance (Light Tolerant)

Feature	Importance
Artificial Light	0.230950
NDVI	0.163174
Temperature	0.020697

Precipitation	0.031523
Seasonality	0.008989
Land Cover	0.000000
Historical Species	0.544666

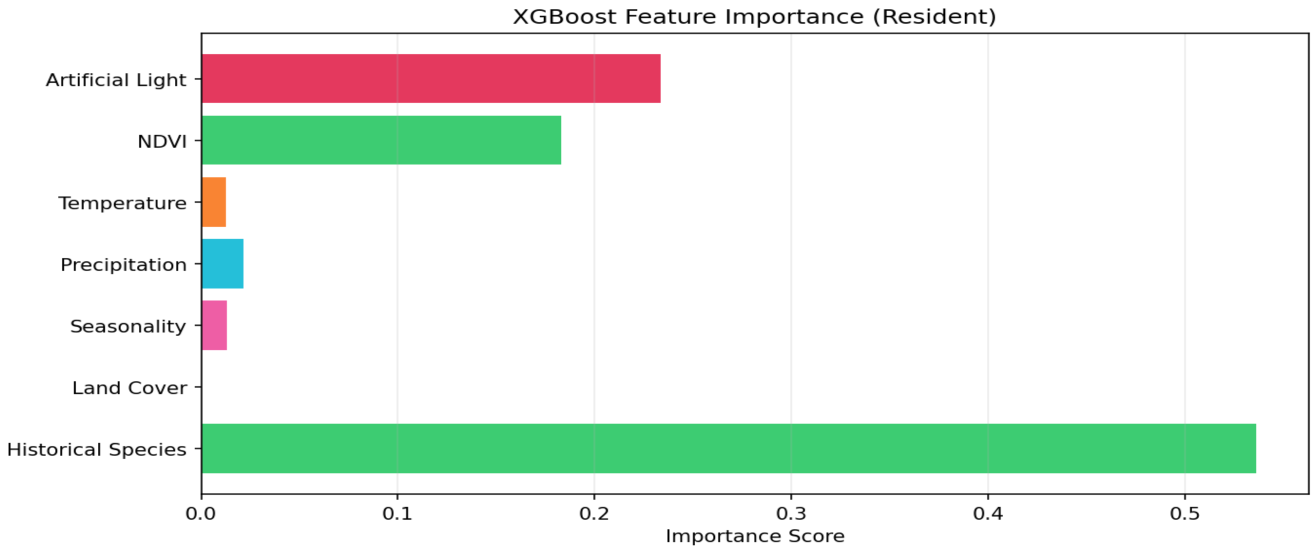
Chart: XGBoost Feature Importance (Migratory)



Numerical Values: XGBoost Feature Importance (Migratory)

Feature	Importance
Artificial Light	0.188284
NDVI	0.034494
Temperature	0.106226
Precipitation	0.040161
Seasonality	0.288632
Land Cover	0.000000
Historical Species	0.342202

Chart: XGBoost Feature Importance (Resident)



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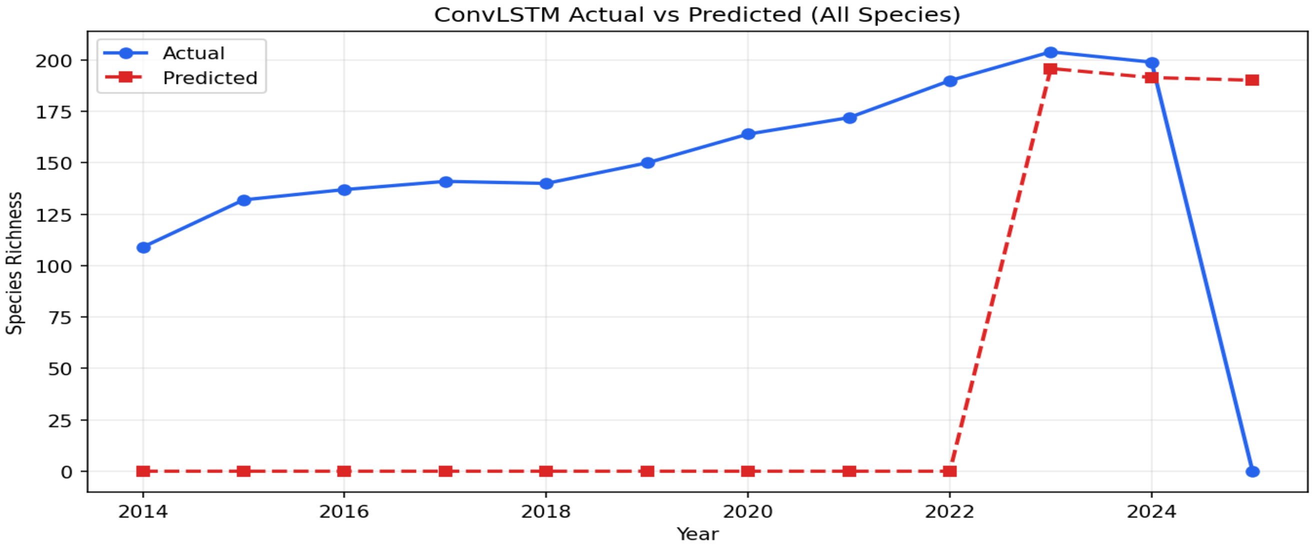
Numerical Values: XGBoost Feature Importance (Resident)

Feature	Importance
Artificial Light	0.233587
NDVI	0.182887
Temperature	0.012712
Precipitation	0.021700
Seasonality	0.012960
Land Cover	0.000000
Historical Species	0.536153

Chart: ConvLSTM Actual vs Predicted (All Species)



Numerical Values: ConvLSTM Actual vs Predicted (All Species)

Year	Actual	Predicted
2014	109.0000	0.0000
2015	132.0000	0.0000
2016	137.0000	0.0000
2017	141.0000	0.0000
2018	140.0000	0.0000
2019	150.0000	0.0000
2020	164.0000	0.0000
2021	172.0000	0.0000
2022	190.0000	0.0000
2023	204.0000	195.9000
2024	199.0000	191.5000
2025	0.0000	190.2000

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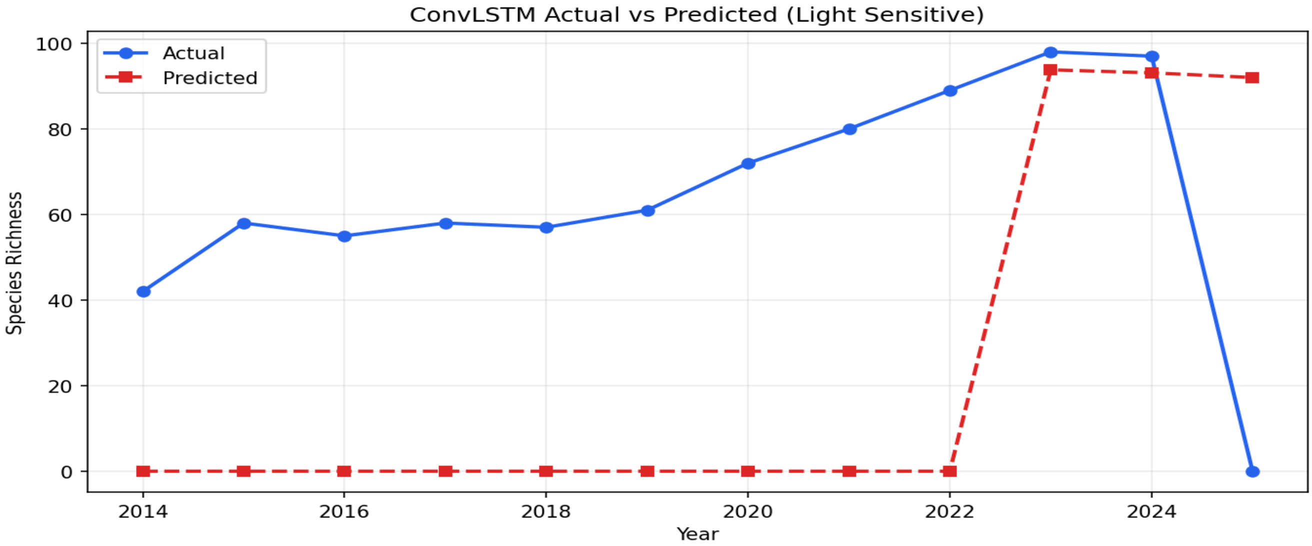
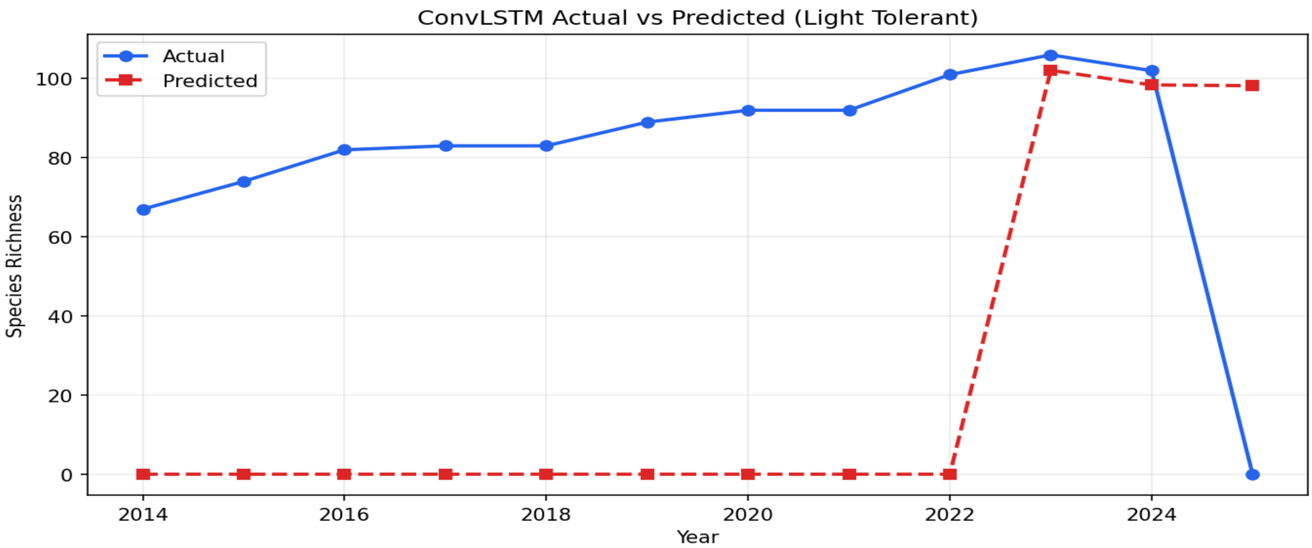
Chart: ConvLSTM Actual vs Predicted (Light Sensitive)



Numerical Values: ConvLSTM Actual vs Predicted (Light Sensitive)

Year	Actual	Predicted
2014	42.0000	0.0000
2015	58.0000	0.0000
2016	55.0000	0.0000
2017	58.0000	0.0000
2018	57.0000	0.0000
2019	61.0000	0.0000
2020	72.0000	0.0000
2021	80.0000	0.0000
2022	89.0000	0.0000
2023	98.0000	93.8000
2024	97.0000	93.1000
2025	0.0000	92.0000

Chart: ConvLSTM Actual vs Predicted (Light Tolerant)



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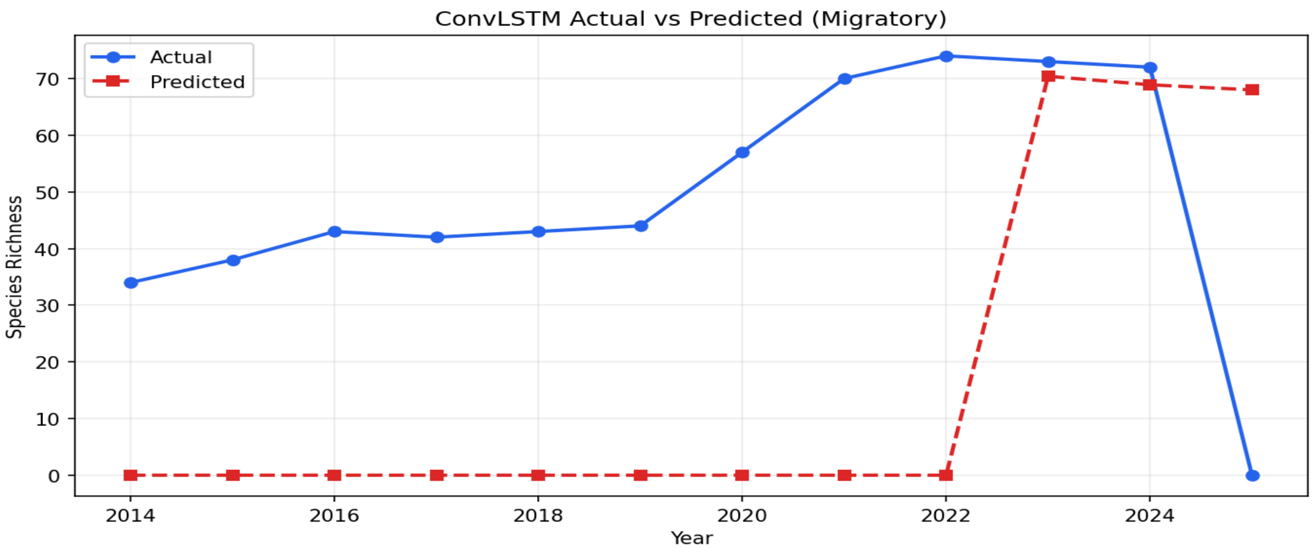
Numerical Values: ConvLSTM Actual vs Predicted (Light Tolerant)

Year	Actual	Predicted
2014	67.0000	0.0000
2015	74.0000	0.0000
2016	82.0000	0.0000
2017	83.0000	0.0000
2018	83.0000	0.0000
2019	89.0000	0.0000
2020	92.0000	0.0000
2021	92.0000	0.0000
2022	101.0000	0.0000
2023	106.0000	102.1000
2024	102.0000	98.4000
2025	0.0000	98.2000

Chart: ConvLSTM Actual vs Predicted (Migratory)



Numerical Values: ConvLSTM Actual vs Predicted (Migratory)

Year	Actual	Predicted
2014	34.0000	0.0000
2015	38.0000	0.0000
2016	43.0000	0.0000
2017	42.0000	0.0000
2018	43.0000	0.0000
2019	44.0000	0.0000
2020	57.0000	0.0000
2021	70.0000	0.0000
2022	74.0000	0.0000
2023	73.0000	70.4000
2024	72.0000	68.9000
2025	0.0000	68.0000

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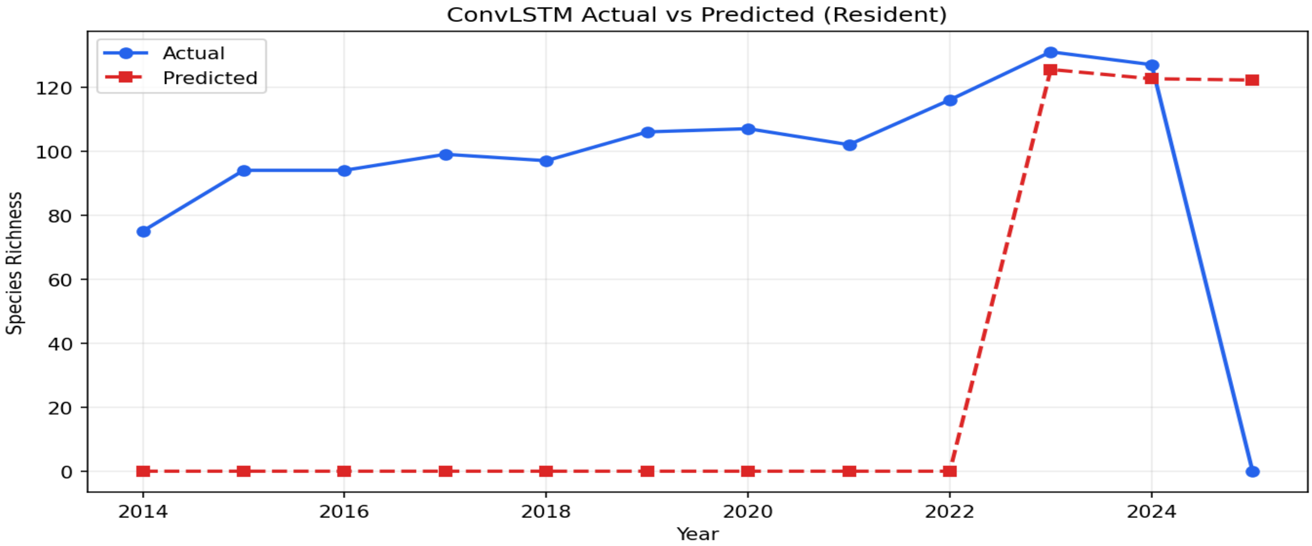
Chart: ConvLSTM Actual vs Predicted (Resident)



Numerical Values: ConvLSTM Actual vs Predicted (Resident)

Year	Actual	Predicted
2014	75.0000	0.0000
2015	94.0000	0.0000
2016	94.0000	0.0000
2017	99.0000	0.0000
2018	97.0000	0.0000
2019	106.0000	0.0000
2020	107.0000	0.0000
2021	102.0000	0.0000
2022	116.0000	0.0000
2023	131.0000	125.5000
2024	127.0000	122.6000
2025	0.0000	122.2000

Conservation Priorities & Recommendations

Area: All Areas | Snapshot range: 2025-01 to 2025-01

Key Biodiversity Area & Protected Areas Audit Table (Core)

Site	Type	Species	Sens.Cnt	Sens.%	Light	Eff.	Status
Manila Bay Beach Resort	PA	12	1	8.3	63.4	24.1	Critical
Luneta National Park	PA	29	6	20.7	78.9	34.3	Critical
Ninoy Aquino Parks and Wildlife Center	PA	41	10	24.4	48.2	46.4	At Risk
Las Piñas-Parañaque Wetland Park	PA	64	21	32.8	14.9	53.1	At Risk
Manila Bay	KBA	92	35	38.0	8.9	55.4	At Risk

Key Biodiversity Area & Protected Areas Audit Table (Environment)

Site	Mean NDVI	Max LST	Precip	Cells	Snapshot
Manila Bay Beach Resort	0.1443	30.23	5186.17	45	2025-12
Luneta National Park	0.2219	27.39	664.79	8	2025-12
Ninoy Aquino Parks and Wildlife Center	0.4275	30.91	991.47	9	2025-12
Las Piñas-Parañaque Wetland Park	0.2009	27.33	4437.85	30	2025-12
Manila Bay	0.2333	30.27	91895.86	874	2025-12

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Rule-Based Recommendations

Reports Tab: Site-Level Decision Path

Source: Reports tab + KBA/PA audit table

Reports tab audit indicates 2 Critical and 3 At Risk sites. Lowest score is 24.1 at Manila Bay Beach Resort. A first intervention queue may focus on Manila Bay Beach Resort, Luneta National Park, Ninoy Aquino Parks and Wildlife Center, with shielded low-CCT lighting considered where VIIRS > 40.0 nW and canopy/cooling interventions suggested where NDVI < 0.22 or LST > 35°C (current counts: light=3, NDVI=2, LST=0). These sites may serve as the first decision-support queue for review and scheduling.

Evidence:

- Weakest sites: Manila Bay Beach Resort, Luneta National Park, Ninoy Aquino Parks and Wildlife Center
- Critical sites: 2
- At Risk sites: 3
- High VIIRS > 40.0 nW: 3
- Low NDVI < 0.22: 2
- High LST > 35.0°C: 0

Night-Light Hotspot Review

Source: Reports tab + KBA/PA audit table

A targeted night-light review may be appropriate for 3 KBA/PA sites with VIIRS above 40.0 nW. Shielding, dimming, or fixture replacement may be considered, with follow-up review in the next monthly cycle.

Evidence:

- High VIIRS sites: 3
- Critical sites: 2
- Threshold used: VIIRS > 40.0 nW

Habitat Recovery and Heat-Buffer Opportunity

Source: Reports tab + KBA/PA audit table

Canopy planting, understory recovery, and heat-buffer surfaces may be suitable in sites with low NDVI or elevated LST. These actions may help improve habitat quality and reduce thermal stress in urban protected areas.

Evidence:

- Low NDVI sites: 2
- High LST sites: 0
- NDVI threshold used: < 0.22
- LST threshold used: > 35.0°C

Dashboard Tab: Multi-Year Ecological Trend Action

Source: Dashboard tab trend summary

Dashboard trend window (2012-2025) shows richness 0.0 ? 55.0 (+55.0), VIIRS 0.0 ? 20.1 (+20.1), and NDVI 0.000 ? 0.345 (+0.345). Quarterly patrols, restoration funding, and compliance review may be aligned to districts where richness is flattening while light is rising.

Evidence:

- Window: 2012-2025
- Richness change: 55.0
- VIIRS change: 20.1
- NDVI change: 0.345

Seasonal Patrol and Monitoring Calendar

Source: Dashboard tab trend summary

The dashboard trend direction may help guide seasonal field patrols and remote sensing reviews. If richness is falling and VIIRS is increasing, inspection frequency may be tightened and outcomes checked after each quarter.

Evidence:

- Trend richness delta: 55.0
- Trend VIIRS delta: 20.1
- Trend NDVI delta: 0.345

Analytics/Diagnostics: Model Reliability Guardrails

Source: Analytics tab diagnostics

Model diagnostics currently report R^2=0.7445, RMSE=6.3924, MAE=4.0568. Model outputs may be treated as triage support, with field validation suggested before high-cost interventions; retraining or recalibration may be considered when R^2 drops below 0.65 or RMSE rises above 9.

Evidence:

- Metrics source: diagnostics_precomputed.json
- R^2: 0.7445
- RMSE: 6.3924

- MAE: 4.0568

Analytics Tab: Driver-Focused Monitoring

Source: Analytics tab feature importance

Feature-importance analysis identifies Artificial Light as the strongest driver (23.1% contribution). Monitoring resources may be allocated to this covariate in hotspot sites, and trigger thresholds in Settings may be updated when monthly drift exceeds baseline.

Evidence:

- Top feature: Artificial Light
- Contribution: 23.1%